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LEARNER GUIDE

For

Microsoft Azure Data Fundamentals (DP-900)

TGS Ref No: TGS-2023036641

Conducted by

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	8 July 2026	Initial DP-900 Learner Guide aligned to all 10 labs, with Markdown mirror and step-by-step activities.	Tertiary Infotech Academy Pte Ltd

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How to Use This Guide

This Learner Guide supports Microsoft Azure Data Fundamentals (DP-900) (TGS-2023036641). Work through the labs in order and use each validation section before moving on.

Course registration: https://www.tertiarycourses.com.sg/wsq-microsoft-azure-data-fundamentals-dp-900.html

Lab repository: https://github.com/tertiarycourses/TGS-2023036641---Microsoft-Azure-Data-Fundamentals-DP-900-Course-Code-TGS-2023036641

Course Details

Item	Details
Course Title	Microsoft Azure Data Fundamentals (DP-900)
Course Code	TGS-2023036641
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN: 201200696W)
Duration	2 days, 16 training hours
Labs	10 guided labs aligned to DP-900 fundamentals
Assessment	Written Assessment and Practical Performance, open book

Learning Outcomes

- Describe structured, semi-structured and unstructured data.
- Compare transactional and analytical workloads.
- Identify common data roles and data store options.
- Explain relational concepts, SQL, normalization and database objects.
- Describe Azure SQL and open-source database services on Azure.
- Describe Azure Storage, Azure Cosmos DB and NoSQL patterns.
- Explain ingestion, processing, analytical data stores, batch and streaming analytics.
- Describe Microsoft Fabric, Azure Databricks, real-time analytics and Power BI concepts.
- Match data scenarios to appropriate Azure data services for DP-900 exam readiness.

Environment Setup

1. Sign in to the Azure portal or Microsoft Learn sandbox provided by your instructor.
2. Open Azure Cloud Shell when CLI commands are required.
3. Use free tiers, trials and sample datasets where possible.
4. Do not upload confidential or personal data into training resources.
5. Clean up resources at the end of class unless the trainer instructs otherwise.

DP-900 Visual Maps

DP-900 Topic Learning Path

Topic 01	Topic 02	Topic 03	Topic 04	Topic 05
Core Data Concepts Labs 1-2	Relational Data Labs 3-4	Non-relational Data Labs 5-6	Analytics Workloads Labs 7-8	Power BI + Exam Readiness Labs 9-10

The following diagrams show how the labs connect the DP-900 concepts into an end-to-end data platform and service-recognition map.

DP-900 Data Platform Flow

Follows the labs: classify data, choose stores, ingest and process data, then model and visualize in Power BI.

Lab-to-Service Map

Lab Guide

Topic 01: Core Data Concepts

Data types, workloads, roles, stores and file formats

Lab 1: Core Data Concepts, Data Types, Workloads

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Distinguish structured, semi-structured, and unstructured data.
- Compare transactional and analytical workloads.
- Identify common data store use cases.
- Build a data concepts glossary.

Scenario:

A retail company stores sales transactions, customer profiles, product images, web logs, invoices, and analytics reports. You must classify the data and decide which workloads are transactional or analytical.

Step-by-step:

1. **Classify Data Types**

Create a table:

Data Example	Type	Reason
Sales order table	Structured	Fixed rows and columns
JSON web event	Semi-structured	Flexible fields with structure
Product photo	Unstructured	Binary media file

Add at least five more examples.

2. **Compare Workloads**

Create a table:

Workload	Purpose	Example
Transactional	Supports day-to-day operations	Online order processing
Analytical	Supports reporting and insight	Monthly sales trend analysis

3. **Identify Workload Characteristics**

For each scenario, choose transactional or analytical:

1. Insert a customer order.
2. Update inventory quantity.
3. Analyze five years of sales data.
4. Create a Power BI dashboard.
5. Stream click events for real-time monitoring.

4. **Create a Glossary**

Define:

Database
Table
File
Data lake
Data warehouse
Transactional workload
Analytical workload

Validation:

You should have a data type table, workload table, classification answers, and glossary.

Checkpoint Questions:

5. What is structured data?
6. What is semi-structured data?
7. Why is unstructured data common in modern systems?
8. How does an analytical workload differ from a transactional workload?

Exam Focus: DP-900 frequently tests the difference between data representations and workload types.

Lab 2: Data Roles, Storage Options, File Formats

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Identify data workload roles and responsibilities.
- Compare common file formats.
- Match storage options to use cases.
- Understand common data store categories.

Scenario:

The retail company is forming a data team. Management asks which people and services are needed for databases, pipelines, analytics, and reports.

Step-by-step:

1. **Identify Data Roles**

Create a table:

Role	Main Responsibility
Database administrator	Manage database availability, performance, backup, and security
Data engineer	Build data pipelines and prepare data for analytics
Data analyst	Build reports, models, and insights

Add examples of tasks for each role.

2. **Compare File Formats**

Format	Common Use
CSV	Simple tabular exchange
JSON	Semi-structured API and event data
Parquet	Columnar analytics data
XML	Structured document exchange
Avro	Data serialization in pipelines

3. **Match Storage Options**

Requirement	Possible Store
Store images and backups	Blob storage
Shared file access	Azure Files
Key-value style entities	Azure Table Storage
Relational application database	Azure SQL Database
Globally distributed NoSQL app	Azure Cosmos DB
Large-scale analytics	Data lake, warehouse, Fabric lakehouse

4. **Draw a Simple Data Platform**

Use diagrams.net:

Source systems -> Ingestion -> Storage -> Processing -> Analytics -> Visualization

5. **Record Service Selection Rules**

Write one sentence for when to use relational, non-relational, object storage, and analytical stores.

Validation:

You should have role notes, file format comparison, storage mapping, and a platform diagram.

Checkpoint Questions:

6. What does a data engineer do?
7. Why is Parquet common in analytics?
8. When would Blob storage be appropriate?
9. Why do analytics platforms separate storage and compute?

Exam Focus: Know the roles and responsibilities for data workloads and common storage formats.

Topic 02: Relational Data on Azure

Relational design, SQL, Azure SQL and open-source databases

Lab 3: Relational Data, SQL, Normalization, Database Objects

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Explain relational data concepts.
- Identify tables, rows, columns, keys, and relationships.
- Understand normalization.
- Recognize common SQL statements and database objects.

Scenario:

The retail company stores customers, orders, products, and order lines in a relational database. You must explain how the data is structured and queried.

Step-by-step:

1. **Draw a Relational Model**

Create entities:

Customers
Orders
OrderItems
Products

Add primary keys and foreign keys.

2. **Explain Normalization**

Answer:

1. Why should product data not be repeated on every order line?
2. How do separate tables reduce update errors?
3. What is the tradeoff of normalization?

3. **Identify SQL Statement Types**

Create a table:

SQL Statement	Purpose
SELECT	Read data
INSERT	Add data
UPDATE	Modify data
DELETE	Remove data
CREATE	Create objects

4. **Review Database Objects**

Define:

Table
View
Index
Stored procedure
Schema
Primary key
Foreign key

5. **Write Example Queries**

```
SELECT ProductName, Price
FROM Products;
```

```
SELECT c.CustomerName, o.OrderDate
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID;
```

Validation:

You should have a relational diagram, normalization notes, SQL table, object definitions, and example queries.

Checkpoint Questions:

6. What is a primary key?
7. What is a foreign key?
8. Why is normalization used?
9. What does SELECT do?

Exam Focus: DP-900 expects recognition of relational concepts and common SQL statements, not advanced SQL programming.

Lab 4: Azure SQL and Open-Source Databases on Azure

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Describe the Azure SQL family.
- Compare Azure SQL Database, Azure SQL Managed Instance, and SQL Server on Azure VMs.
- Identify open-source database services on Azure.
- Match relational services to scenarios.

Scenario:

The retail company wants to migrate existing relational databases to Azure. Some apps use SQL Server, while others use PostgreSQL and MySQL.

Step-by-step:

1. **Compare Azure SQL Options**

Service	Best Fit
Azure SQL Database	Modern cloud app needing managed PaaS database
Azure SQL Managed Instance	SQL Server compatibility with managed service benefits
SQL Server on Azure VMs	Maximum OS and instance-level control

2. **Compare Open-Source Database Services**

Document:

Azure Database for PostgreSQL
Azure Database for MySQL
Azure Database for MariaDB legacy considerations

3. **Match Migration Scenarios**

Choose a service:

1. New cloud-native app using SQL Server engine.
2. Existing SQL Server app needing instance-level compatibility.
3. Legacy app requiring OS-level control.
4. Web app using PostgreSQL.
5. App using MySQL.

4. **Review Managed Service Benefits**

Write how managed database services help with:

- Patching.
- Backup.
- High availability.
- Scaling.
- Monitoring.
- Security.

5. **Security and Cost Notes**

Document:

Authentication
Firewall or private access
Backup retention
Performance tier
Scaling choice

Validation:

You should have comparison tables, scenario answers, managed service benefits, and security/cost notes.

Checkpoint Questions:

6. When is Azure SQL Database appropriate?
7. Why choose SQL Server on Azure VMs?
8. What is a managed database service?
9. Which Azure services support PostgreSQL and MySQL?

Exam Focus: Know the Azure relational database service families and when each option fits.

Topic 03: Non-Relational Data on Azure

Azure Storage, Blob, Files, Tables and Cosmos DB

Lab 5: Azure Storage, Blob, Files, Tables

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Describe Azure Blob Storage.
- Describe Azure Files.
- Describe Azure Table Storage.
- Match storage services to scenarios.
- Review access tiers and redundancy concepts.

Scenario:

The retail company needs to store product images, shared documents, application logs, and simple key-value data.

Step-by-step:

1. **Map Storage Needs**

Requirement	Azure Storage Service
Product images	Blob Storage
Shared file share	Azure Files
Simple key-value customer	Table Storage

preferences	
Queue messages	Azure Queue Storage

2. **Review Blob Storage Concepts**

Define:

Container
Blob
Block blob
Access tier
Lifecycle management

3. **Review Azure Files Concepts**

Explain:

- SMB and NFS file shares.
- Lift-and-shift file server scenarios.
- Shared access for applications.

4. **Review Table Storage Concepts**

Explain:

- Partition key.
- Row key.
- Entity.
- Schemaless design.

5. **Compare Redundancy**

Write a short note on locally redundant, zone-redundant, and geo-redundant storage.

Validation:

You should have storage mappings, concept definitions, and redundancy notes.

Checkpoint Questions:

6. When should you use Blob Storage?
7. When should you use Azure Files?
8. What is Azure Table Storage?
9. Why does redundancy matter?

Exam Focus: DP-900 expects recognition of Azure Storage services and their common use cases.

Lab 6: Azure Cosmos DB, NoSQL, APIs

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Explain non-relational data concepts.

- Describe Azure Cosmos DB capabilities.
- Identify common Cosmos DB APIs.
- Match NoSQL patterns to scenarios.

Scenario:

The retail company wants a globally distributed product catalog and user profile store that can handle flexible schemas and low-latency reads.

Step-by-step:

1. **Explain NoSQL Fit**

Write why NoSQL can help with:

- Flexible schema.
- High scale.
- Global distribution.
- Low-latency access.
- JSON document data.

2. **Review Cosmos DB Capabilities**

Document:

Global distribution
Low latency
Throughput with request units
Partitioning
Consistency levels
Multiple APIs

3. **Identify Cosmos DB APIs**

Create a table:

API	Common Fit
NoSQL	JSON document applications
MongoDB	MongoDB-compatible apps
Cassandra	Wide-column workloads
Gremlin	Graph workloads
Table	Key-value/table workloads

4. **Design a Product Catalog Document**

Create example JSON:

```
{
  "id": "P1001",
  "category": "Laptop",
  "name": "Training Laptop",
  "price": 1200,
```

```
  "tags": ["business", "portable"]
}
```

5. **Review Partitioning**

Explain why a good partition key distributes data and workload evenly.

Validation:

You should have NoSQL notes, Cosmos DB capability notes, API mapping, sample document, and partition key explanation.

Checkpoint Questions:

6. When is Cosmos DB a good fit?
7. What is a request unit?
8. Why does partitioning matter?
9. Which API supports graph workloads?

Exam Focus: Know Cosmos DB capabilities, APIs, and common NoSQL use cases.

Topic 04: Analytics Workloads on Azure

Ingestion, processing, analytical stores, Fabric, Databricks and real-time analytics

Lab 7: Analytics, Ingestion, Processing, Analytical Data Stores

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Describe analytics workload stages.
- Explain data ingestion and processing.
- Compare batch and streaming.
- Identify analytical data stores.
- Map services to analytics scenarios.

Scenario:

The retail company wants daily sales reports, customer behavior analytics, and near-real-time alerting from web clickstream data.

Step-by-step:

1. **Map an Analytics Pipeline**

Create a flow:

Data sources
Ingestion
Raw storage
Transformation
Analytical store

Semantic model
Visualization

2. **Compare Batch and Streaming**

Mode	Description	Example
Batch	Processes data in groups on a schedule	Daily sales summary
Streaming	Processes events continuously	Live website activity

3. **Identify Analytical Stores**

Document:

Data lake
Data warehouse
Lakehouse
Semantic model

4. **Choose Azure Services**

Need	Possible Service
Data ingestion and orchestration	Data Factory or Fabric Data Factory
Big data processing	Azure Databricks
Lakehouse analytics	Microsoft Fabric
Real-time event processing	Event Hubs, Stream Analytics, Fabric Real-Time Intelligence
Visualization	Power BI

5. **Review Data Transformation**

Explain the difference between raw, cleaned, curated, and aggregated data.

Validation:

You should have pipeline diagram, batch/streaming comparison, analytical store notes, and service mapping.

Checkpoint Questions:

6. What is data ingestion?
7. What is the difference between batch and streaming?
8. What is a data warehouse?
9. What is a lakehouse?

Exam Focus: DP-900 analytics questions test high-level pipeline patterns and service selection.

Lab 8: Microsoft Fabric, Azure Databricks, Real-Time Analytics

Objectives	Guided Activities	Learner Output	Exam Focus

Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Describe Microsoft Fabric at a high level.
- Describe Azure Databricks use cases.
- Identify real-time analytics services.
- Match analytics services to scenarios.

Scenario:

The retail company is comparing analytics platforms for reporting, data engineering, machine learning, and real-time dashboards.

Step-by-step:

1. **Review Microsoft Fabric Concepts**

Document:

OneLake
Lakehouse
Warehouse
Data Factory
Real-Time Intelligence
Power BI

2. **Review Azure Databricks Concepts**

Explain how Databricks can support:

- Apache Spark processing.
- Data engineering notebooks.
- Machine learning workloads.
- Delta Lake patterns.
- Collaborative analytics.

3. **Identify Real-Time Analytics**

Match:

Requirement	Possible Service
Ingest event streams	Event Hubs
Process streaming data	Stream Analytics
Real-time analytics in Fabric	Real-Time Intelligence
Visualize live metrics	Power BI real-time dashboards

4. **Compare Services**

Create a decision table:

Scenario	Better Fit
Business reporting workspace	Microsoft Fabric
Spark-heavy engineering workloads	Azure Databricks
Simple stream processing	Azure Stream Analytics
Event ingestion at scale	Event Hubs

5. **Draw an Analytics Architecture**

Point of sale -> Event stream -> Real-time processing -> Dashboard
Daily exports -> Lakehouse -> Transformations -> Power BI report

Validation:

You should have Fabric notes, Databricks notes, real-time service mapping, and architecture diagram.

Checkpoint Questions:

6. What is Microsoft Fabric?
7. What is Azure Databricks commonly used for?
8. What is real-time analytics?
9. Which service can ingest event streams?

Exam Focus: Know the high-level purpose of Fabric, Databricks, and Microsoft real-time analytics services.

Topic 05: Power BI and Exam Readiness

Models, visualizations, service mapping and capstone review

Lab 9: Power BI, Data Models, Visualizations

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Describe Power BI capabilities.
- Explain data models and semantic models.
- Choose appropriate visualizations.
- Build a simple report plan.

Scenario:

The retail company wants a dashboard for sales, products, regions, and trends. You must explain how Power BI supports data analysis.

Step-by-step:

1. **Identify Power BI Components**

Document:

Power BI Desktop
Power BI service
Dataset or semantic model
Report
Dashboard
Workspace
Data refresh

2. **Design a Simple Data Model**

Create tables:

Sales
Products
Customers
Dates
Regions

Identify relationships and keys.

3. **Choose Visualizations**

Question	Visualization
Sales by month	Line chart
Sales by product category	Bar chart
Sales by region	Map or filled map
Market share by category	Donut chart or treemap
Detailed sales records	Table

4. **Add Measures and Filters**

Define examples:

Total Sales
Order Count
Average Order Value
Year
Region
Product Category

5. **Review Good Dashboard Design**

Write five rules for clear dashboards, such as consistent labels, useful filters, and avoiding clutter.

Validation:

You should have Power BI component notes, data model design, visualization choices, and dashboard rules.

Checkpoint Questions:

6. What is Power BI used for?
7. What is a semantic model?
8. Why are relationships important?
9. Which visualization is suitable for trends over time?

Exam Focus: Power BI questions test capabilities, data models, and choosing appropriate visualizations.

Lab 10: DP-900 Capstone and Exam Readiness

Objectives	Guided Activities	Learner Output	Exam Focus
Read what the lab is meant to prove.	Complete each numbered task in order.	Keep the tables, notes, diagrams and answers.	Use the checkpoint questions for review.

Objectives:

- Review all DP-900 skill areas.
- Design an end-to-end Azure data platform.
- Match data scenarios to Azure services.
- Build a personal exam study plan.
- Clean up lab resources.

Scenario:

The retail company asks for a data platform recommendation that supports transactions, product images, customer profiles, analytics, real-time events, and dashboards.

Step-by-step:

1. **Build a Service Map**

Requirement	Data Type or Workload	Azure Service
Online orders	Transactional relational	Azure SQL Database
Product images	Unstructured object data	Blob Storage
Shared files	File storage	Azure Files
Flexible product catalog	NoSQL document data	Azure Cosmos DB
Daily analytics	Analytical workload	Fabric lakehouse or warehouse
Event stream	Streaming workload	Event Hubs or Stream Analytics
Reports	Visualization	Power BI

2. **Draw the Architecture**

Include:

Transactional applications
Operational databases
Storage accounts
Ingestion pipelines
Analytical store
Power BI reports
Real-time event path

3. **Review DP-900 Skill Areas**

Rate confidence:

Skill Area	Confidence 1-5	Study Action
Core data concepts		
Relational data on Azure		
Non-relational data on Azure		
Analytics workloads on Azure		

4. **Build an Exam Mistakes Log**

For missed questions, record:

Topic:
Wrong assumption:
Correct concept:
Service to review:

5. **Clean Up Resources**

If you created resources:

az group delete --name rg-dp900-lab --yes --no-wait

Confirm with your instructor before deleting shared resources.

6. **Create a 7-Day Study Plan**

1. Day 1: Core data concepts and workload types.
2. Day 2: Relational concepts and SQL.
3. Day 3: Azure SQL and open-source databases.
4. Day 4: Azure Storage and Cosmos DB.
5. Day 5: Analytics pipelines and data stores.
6. Day 6: Fabric, Databricks, real-time analytics, Power BI.
7. Day 7: Practice assessment and mistakes log review.

Validation:

You should have a service map, architecture diagram, confidence matrix, study plan, and cleanup confirmation.

Checkpoint Questions:

7. Which Azure data service is easiest for you to identify?

8. Which service names are most confusing?
9. What is the difference between transactional and analytical data?
10. What will you review before exam day?

Exam Focus: DP-900 rewards service recognition and concept clarity. Focus on what each data service is for, not deep administration tasks.

Assessment Flow

11. TRAQOM: scan the QR code on the LMS.
12. Take Assessment digital attendance.
13. Complete Written Assessment and Practical Performance.
14. Submit answers on the LMS.
15. Sign the Assessment Summary Record.